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Accelerating Product and Service Carbon Footprint Estimations Using a Data Science Approach

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Abstract

This paper presents a data science focused methodology for calculating carbon footprints for products and services in the upstream oil and gas service industry. Complementing traditional approaches, the method leverages existing corporate data sources, machine learning, and other data science techniques to navigate the complexities and challenges associated with conducting life cycle assessments (LCAs) across the extensive range of products and services in this industry, enabling the acceleration of decarbonization actions.

Our approach leverages a comprehensive corporate greenhouse gas inventory (including scopes 1, 2 and 3 both upstream and downstream), along with revenue and associated product and service parameters recorded during delivery to customers. These datasets are integrated, and data-science methods are employed to explore and refine relationships among these parameters. Using data science methodologies, we derive over 561,000 emissions factors for various combinations of countries, products, service types, and timelines. These emissions factors are then further refined using additional techniques to estimate the carbon footprint of individual products and services provided to exploration and production (E&P) customers.

The methodology has been successfully applied to estimate greenhouse gas (GHG) emissions associated with product and service delivery in the E&P service sector, and its results have been validated against cases where domain-specific LCA calculation methodologies and corresponding activity data are available. Although the intention is to complement an LCA approach rather than replace it, our novel method enabled estimations of product and service carbon footprints in a far shorter timeframe and at greater scale. This leads to rapid insights such as emissions trends, hotspots and reduction opportunities both at large scale, such as at country or business unit level, and at small scale, such as for individual operations or product deliveries. It enables targeted insights into what services and products warranted a more traditional LCA approach to characterize their footprint. By reducing the time needed for emissions calculations and linking emissions sources with known decarbonization technologies, this method can accelerate decarbonization actions. Although this paper focuses on a use case specific to the oil and gas service industry, this methodology also has the potential to benefit the broader E&P sector as well as other industries in moving from emissions calculations to decarbonization actions.

Introduction

Product carbon footprint represents the total greenhouse gas (GHG) emissions generated across a product's life cycle, from raw material extraction to end-of-life. Expressed in carbon dioxide equivalents (CO₂e), it accounts for all major sources of GHGs using life cycle thinking to identify emission "hotspots" and support comparisons, reporting, and reduction strategies (ISO, 2018; WRI & WBCSD, 2001).

The underlying methodology has developed over the past three decades, starting with *Life Cycle Assessment* principles in the 1990s (ISO, 1997). The 2000s saw the launch of the *Greenhouse Gas Protocol* (WRI & WBCSD, 2001) and the UK's *PAS 2050* (BSI, 2008), which set consistent rules for product-level accounting. These were later harmonized through ISO 14067 (ISO, 2018) and enhanced by better emission data, sector guidance, and digital tools.

Many oil & gas companies and service companies have set net zero or other emissions reduction targets, and measure and report their corporate GHG footprint each year. Sources of GHG emissions in the oil and gas industry typically include energy consumption-related emissions, carbon dioxide (CO₂) and methane (CH₄) from routine flaring operations and direct CH₄ emissions from both vented and fugitive sources. Direct methane emissions are a significant contributor, representing 47% of all CO₂e emitted within the oil and gas value chain (IEA – 2023). The choice of which emissions source to target varies depending on many factors, but in almost all cases boils down to economics. Which source can I reduce or eliminate in the most cost-effective manner? With such a complex value chain, and so many technological options to choose from, how do I choose the most effective technology for my most significant emission sources? Whilst marginal abatement cost curve (MACC) analysis and ever more sophisticated software solutions can help with this, calculating (and trying to reduce) the footprint of an individual product is emerging as a key driver for targeted decarbonization efforts.

Product carbon footprints (PCFs) can play a pivotal role in reducing emissions in upstream oil and gas by making the climate impact of individual products and activities transparent and quantifiable. By breaking down emissions across the full life cycle—from exploration and drilling to processing and transport—PCFs reveal high-intensity "hotspots" enabling targeted interventions for emissions reduction.

However, product carbon foot printing is not yet deployed at scale within the oil & gas industry. There remains a significant gap in methodology and granularity because much of the emissions data measured today does not include granularity at the product and service level (e.g., product and service level carbon footprint estimations are often not available). Whilst corporate emissions reporting is relatively widespread among major service providers, product carbon footprint calculations are relatively less mature within the industry. There are several good reasons for this. Performing PCFs in the oil and gas service industry is particularly challenging due to the scale and complexity of operations. Service providers often need PCFs for thousands of products and services, creating a requirement for assessment at scale that traditional, manual life cycle assessment methods—often taking several weeks per product—cannot feasibly meet. The industry's extensive portfolios include both manufactured goods and complex field services, each with unique supply chains and operational profiles. Adding to this complexity, products are manufactured, assembled, and delivered across widely varying geographies, meaning that emission factors, transport modes, and energy mixes can differ dramatically for the same item depending on where and how it is produced and deployed. This combination of high volume, time-intensive calculation, and geographic variability makes generating accurate, timely PCFs a significant operational challenge.

So, how can we address the challenge of scaling product and service carbon footprints in the oil and gas service sector and beyond, to help other sectors with a similar level of complexity?

This paper proposes a scalable data science driven approach that integrates enterprise resource planning (ERP) and emissions data to generate estimates of the greenhouse gas emissions at the product and services level, which in turn can enable more targeted emissions reduction plans.

Methodology

The basic premise of the methodology is that a product and/or service specific GHG emissions estimate can be made by combining a corporate emissions database, which includes emissions data at a less granular level than the individual product or service, with enterprise business system data in which we have other data (such as revenue or other operating parameters) which are at the same level of granularity as individual products and/or services. Using a data science focused approach, both sources of data can then be combined to give GHG insights at a more granular level than exists within a typical GHG corporate inventory – as shown here in [Figure 1: Methodology High Level Overview](#)

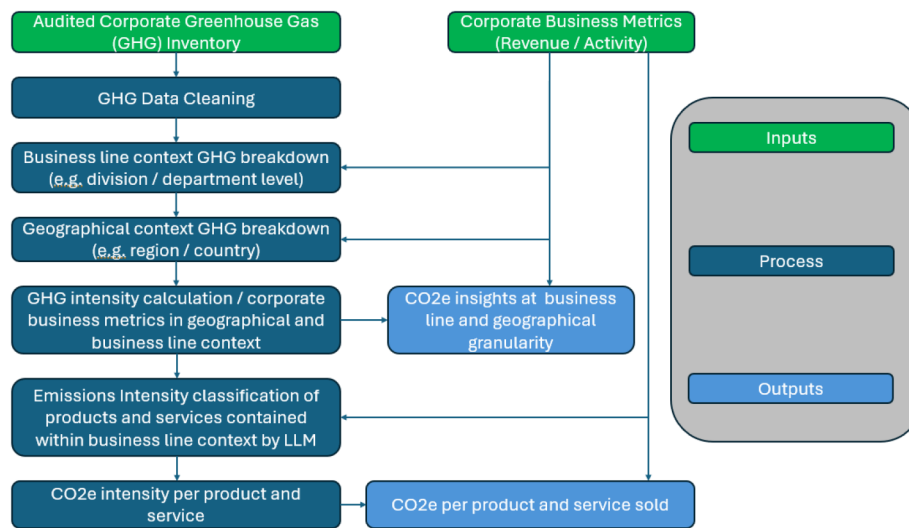


Figure 1—Methodology High Level Overview

The step-by-step methodology is explained in more detail below:

Data Inputs and Cleaning

Since the recording of business metrics is generally more mature than GHG emissions, the level of granularity of these metrics is generally greater than the granularity level of the GHG emissions, and in many cases the metrics are available at the product and/service level. Once the reference metrics are identified, their correlation to GHG emissions can be validated. In our case, revenue was used as the reference metric, and it showed a significant correlation to GHG emissions across all scope categories, regions and business lines across all 4 years considered. Spend data was also analyzed but eventually was not used as a reference metric in our case. See [Figure 2: GHG Emissions, Revenue and Spend Correlation](#)

To be used for further analysis, some data cleanup was required for the GHG data. This mostly involved the removal of any special characters used, for example, unification of country codes and the deletion of redundant columns.

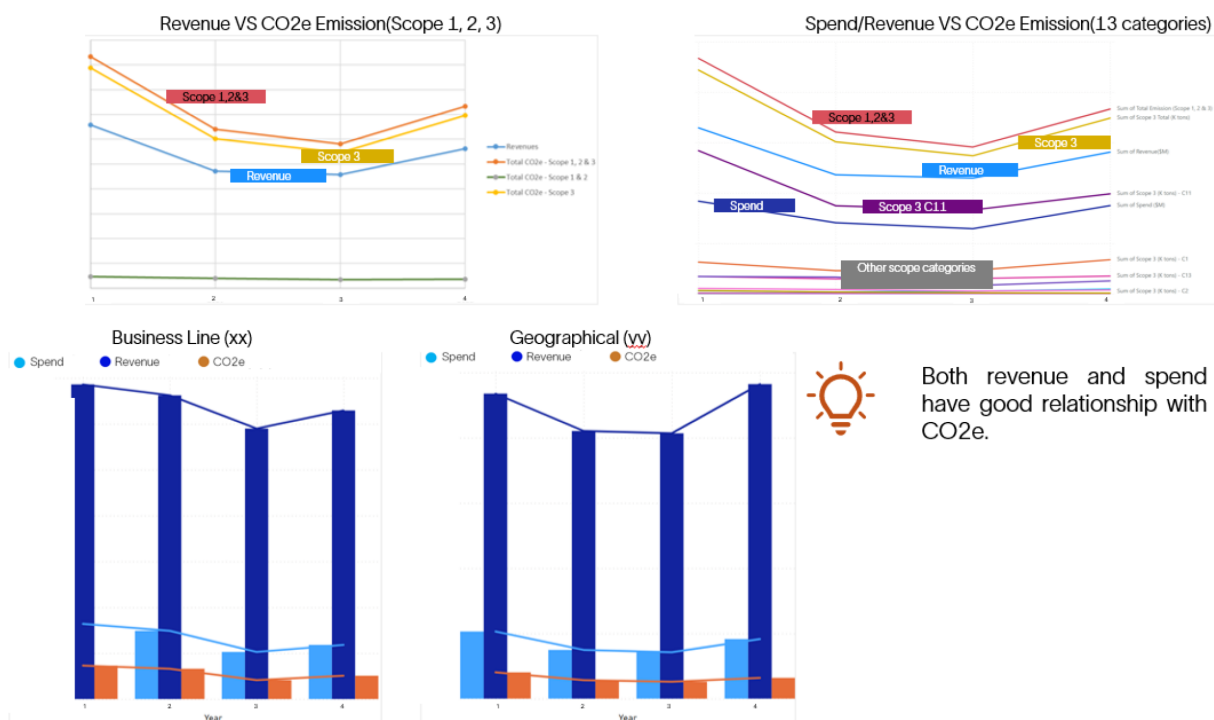


Figure 2—GHG Emissions, Revenue and Spend Correlation

Sub-Business line and Geographical GHG breakdown

This step allocates the GHG emissions down to an intermediate level of granularity, to enable insights into GHG emissions sources and trends in the business and geographical context. In this example, the business context chosen is the sub-business line and geographical context that of the sub-geounit. The sub-business line represents a highly granular level of business organization, responsible for a specific set of products and services, whereas the sub-geounit represents the most granular level of geographical organization, typically corresponding to an individual country. Before the imputation / allocation process, the level of completeness of GHG data at this level of granularity is validated and analysed as indicated below in Figure 3

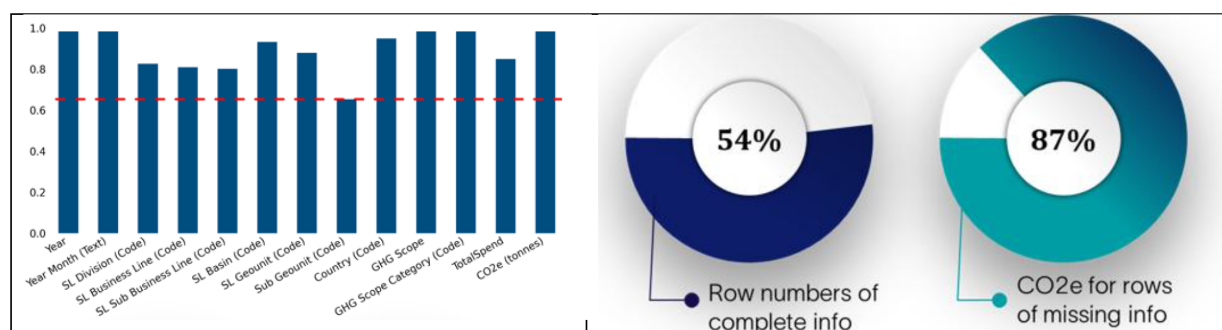


Figure 3—Data Statistics: Rows and Carbon percentage

The chart on the left of Figure 3: Data Statistics: Rows and Carbon percentage indicates the total number of rows with valid GHG emissions data across different parameters present in the corporate GHG emissions database for the four years used for analysis, prior to the allocation / imputation work. The chart on the right in Figure 3 indicates the % quantity of total rows and total GHG emissions for which all the parameters on the left are recorded. Imputation, as described later in this section, is used to fill the gaps, based upon statistical analysis of emissions (where present) and revenue data.

To address gaps in the data, an imputation methodology is applied. GHG emissions contained in the corporate database are first disaggregated at the business line level using distribution features. In this context, the quality of business line information within the GHG corporate inventory surpasses that of the geographic organization data, so imputation for business lines is conducted prior to geographic breakdown.

The imputation process relies on carbon distribution features, which involve calculating the statistical likelihood of GHG emissions across various business lines and geographic organizations at a defined level of granularity. Each possible business line and geographic region is assigned a share of the carbon emissions based on these probabilities, ensuring that the total allocation sums to 100%. Distribution features are calculated according to imputation scenarios, derived either from actual carbon emissions (CO₂e) or, where such data is insufficient, from revenue data. Where possible, emissions data is prioritized for this calculation; revenue is only used when emissions data for a particular business or geographic context is not robust enough for statistical analysis.

Given the inherent limitations within the GHG dataset—specifically the frequent absence of emissions data at the desired levels of detail—we employ imputation as a key strategy to bridge these data gaps. In practice, this approach addresses three primary scenarios in both business and geographical contexts, as outlined below:

Business line context:

- Case 1: GHG data at the division and business line levels are known while sub-business line is missing.
- Case 2: GHG data at division level is known while business line and sub-business line are missing.
- Case 3: All GHG data for division, business line and sub-business are missing.

Key assumptions & logic for business line breakdown:

- For those rows with missing business line organization information, the distribution features should be similar to the features of rows with complete organization information.
- For specific scope/category, if the distribution features with complete organization information were not available, the distribution features from the global data were applied for business line breakdown.
- The business line distribution features were constrained within the relevant geographic organization to ensure relevance.

Geographical context:

- Case 1: Only sub-geounit information is missing.
- Case 2: Both geounit and sub-geounit information are missing.
- Case 3: All basin, geounit and sub-geounit information are missing.

Key assumptions and logic for the geographical breakdown:

- For those rows with missing geographic organization information, the distribution features should be similar as the features of rows with complete organization information.
- For specific scope/category, if the distribution features with complete organization information are not available, some constraints from bottom-up data or global data like revenues were applied

As described, the business line and geographic breakdown are largely similar. The three cases for the geographic breakdown are shown below. [Figure 4](#) CO₂e(S3C1) Propagation with CO₂e Distribution Features for Case 1 and [Figure 5](#) CO₂e(S3C1) Propagation with CO₂e Distribution Feature for Case 2

show examples where the distribution features were computed using CO2e emissions, for specific GHG emissions scope categories. Scope 3 category 1 (S3C1) represents to upstream scope 3 emissions from purchased goods and services, whilst Scope 3 category 4 (S3C4) refers to scope 3 emissions from upstream transportation and distribution (from the GHG Protocol) . The example shown in Figure 6 CO2e(S3C4) Propagation with Revenue Distribution Feature for Case-3 shows a simulated case where revenue is used as a distribution feature.

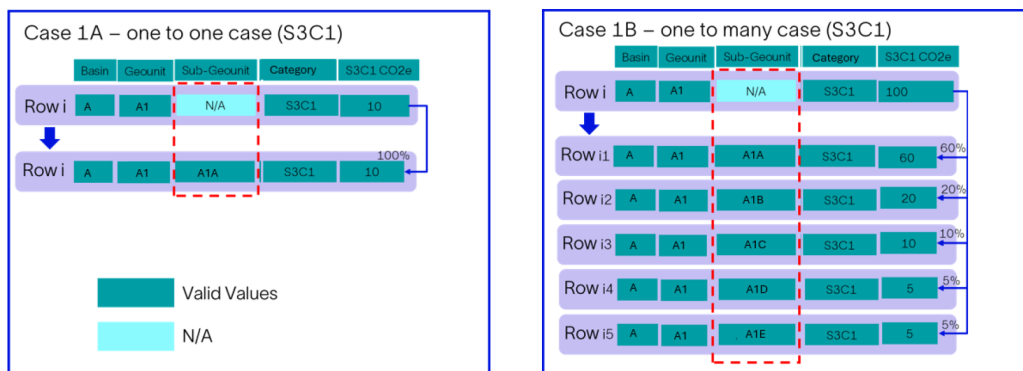


Figure 4—CO2e(S3C1) Propagation with CO2e Distribution Features for Case 1 - for a simulated geographical organization basin A which comprises geounit A1 and sub-geounits A1A – A1E.

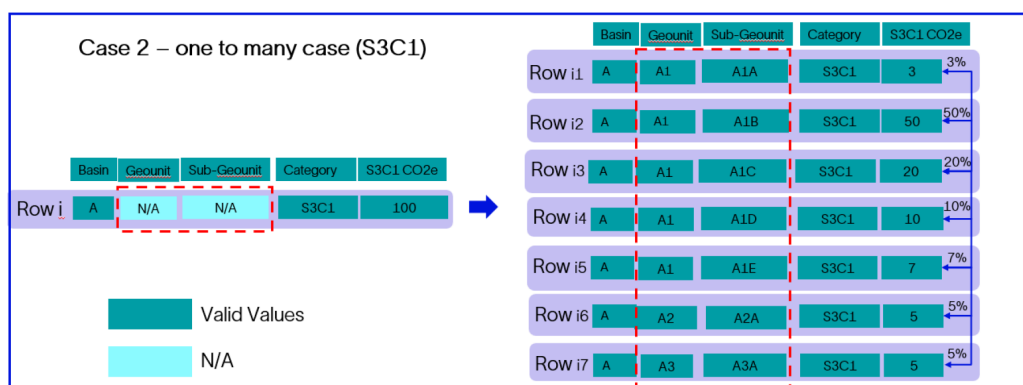


Figure 5—CO2e(S3C1) Propagation with CO2e Distribution Features for Case 2, - for a simulated geographical organization of basin A comprising geounits A1-A3 and sub-geounits A1A – A1E, A2A and A3A

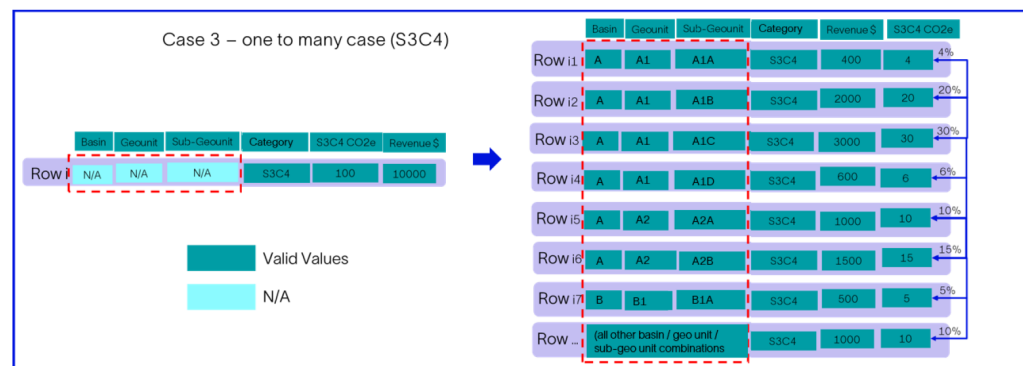


Figure 6—CO2e Propagation with Revenue Distribution Features for Case-3 - - for a simulated geographical organization of basins A and B, geounits A1, A2 and B1 and sub-geounits A1A – A1D, A2A, A2B and B1A.

GHG Intensity Calculations and Customer Specific GHG Emissions Outputs

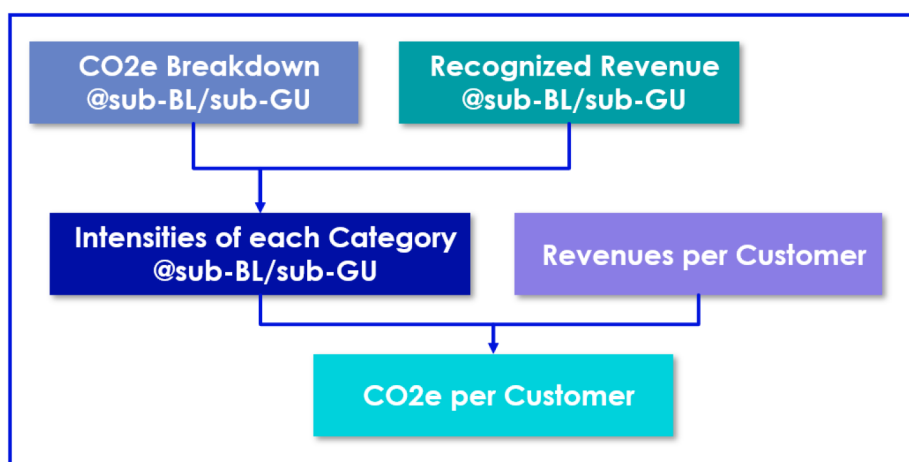
The workflow as described up to this point is useful in adding granularity to a corporate GHG inventory. However, how can this data be used to give insights related to product and service specific GHG emissions data to customers who are purchasing the products and services in question?

The workflow below shows how to calculate the carbon emission for each customer of the oil and gas service company in question. After the imputation process is complete for carbon emissions data in both business and geographical contexts, the carbon emission data for each business line and geounit can be joined with corresponding corporate metrics that were previously determined to correlate to GHG emissions, which in this case is corporate revenue. The "fusion" of these two datasets is enabled since both have a complete population of statistics (emissions by scope category for one dataset and revenue for the other) for all geographical and business line parameters that were considered for the previous imputation process. After merging the two datasets, the GHG data and recognized revenue data can be used to calculate GHG emissions intensities in CO₂e / unit revenue. For a given year, individual GHG emissions intensity factors in T CO₂e/\$ are calculated, in each GHG scope category, for each sub-business line in each sub-geounit. The total output is over 560,000 emissions intensity factors specific to the oil and gas service companies product and service delivery.

The carbon intensities (CO₂e/unit revenue) for each category, are calculated using revenue and carbon emission data as described above. Then the carbon emissions resulting from sales of SLB products and services to each customer can be calculated based upon the specific revenue associated with that customer across a specific time period.

Key Assumptions:

- Intensities of each category are the same for all customers within the same sub-business line and country.
- Within a specific sub-business line and country, carbon emissions for all scope categories are linearly proportional to revenue.



The flow of emissions between different parameters is then visualized using a Sankey chart – see [Figure 10](#)

Carbon Life-cycle Classification by LLM, using Sub-BL level GHG emissions as calculated in above phases

Traditional life cycle analysis in the oil and gas industry involves manual data gathering and analysis, leading to a significant investment of time and resources. Semantic analysis with the help of large language models

(LLM) has become much more popular and has started to apply many scenarios. Leveraging the capabilities of large language models (LLM), particularly exemplified by ChatGPT (OpenAI, 2025) makes it possible to generate intricate life cycle analysis maps. The large language model reduces manual data processing, providing a more efficient and accurate assessment process. For this, we first trained the large language model using the various main stages and processes, described in the OPGEE model "model coverage" section via prompt engineering to use a text-generation function to generate a canonical life-cycle structure matching the oil & gas operational stages and processes described in the same OPGEE model. This life cycle structure can also be modified by the users later. With the canonical structure generated, the LLM model was trained using SLB internal data in how to map the activities or business units to life cycle classes through the natural language processing technique, few-shot prompting. All these procedures were deployed in a Dataiku workflow in a Python recipe approach.

Key Assumptions

- LLM Classification: The business activities within the same sub-BL are similar. The sub-BL represents the concrete activities for the input to LLM classification process. (Can be replaced in future with concrete activities if activities detail available).

The flow chart for the application of LLM for oil & gas value chain analysis is shown in Figure 7: Flowchart of LLM application for Life-Cycle Analysis

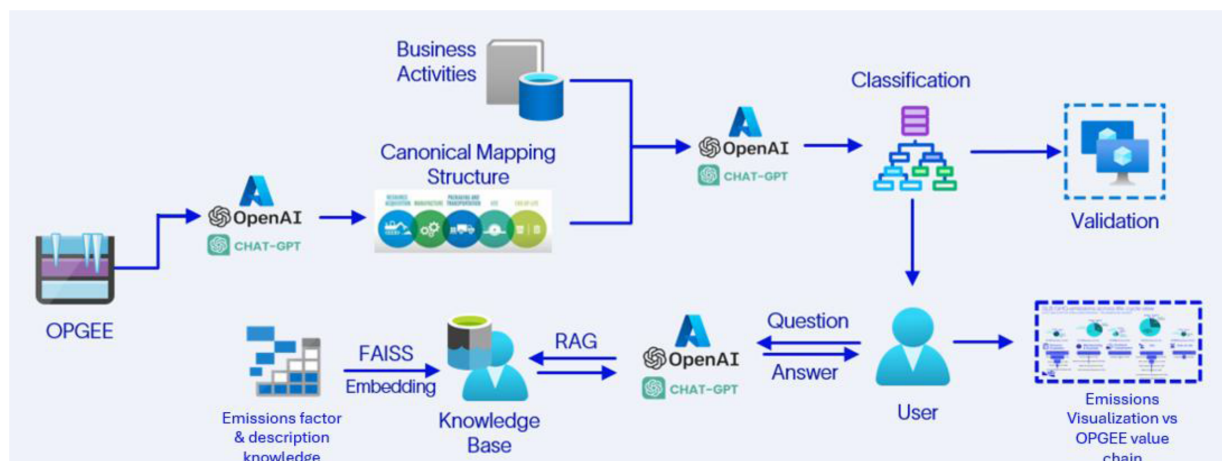


Figure 7—Flowchart of LLM application for Life-Cycle Analysis

For reference, an example of life-cycle stages and processes described within the OPGEE model are shown below:

Table 1—OPGEE stages, processes and sub-processes from OPGEE model 3.0 (Brandt, A. R., et al. (2022))

Main stage	Process	Sub-process	Emissions source
Production and extraction	Lifting	Pumping	Combustion for pump driver
			Electricity for pump driver
			Casing and wellhead fugitive emissions
		Gas lift	Compressor prime mover emissions
			Compression electricity emissions
			Casing and wellhead fugitive emissions
	Injection	Gas injection	External gas processing (e.g., N ₂ production)
			Gas compression energy
			- Gas sequestration credit (CO ₂ flood)
		Steam injection	Water pre-treatment
			OTSG fuel combustion
			Turbine gas consumption (combined cycle)
			HRSG duct firing (combined cycle)
			- Electricity co-production offsets (combined cycle)
			Steam pumping energy (if any)
		Polymer flood	Polymer embodied energy
			Polymer mixing
			Polymer/water mixture pumping energy
		Surfactant/other injection	Surfactant/other embodied energy
			Surfactant/other mixing
			Surfactant/other mixture pumping energy

Product and Service Emissions Intensity Calculations using large language models (LLMs)

Taking the sub-business line specific GHG emissions outputs and further refining these down to the product or service level represents a particular challenge. Without performing a full PCF analysis, how do you come up with reference data to inform an actionable emissions allocation to this level of granularity?

The approach proposed matches descriptions of products and services with known indicators of relative GHG emissions intensity. Two datasets were proposed for this analysis:

1. Third party GHG emissions factors already used to compute the corporate GHG inventory
2. Emissions intensities as recorded against different oil & gas operational stages as defined within The Oil Production Greenhouse Gas Emissions Estimator (OPGEE).

The key challenge in this case was matching product and service descriptions to the correct emissions factor from a library of third-party emissions factors, or to the OPGEE definitions of E&P operational stages. There are more than 23,000 different products and services considered, so a manual match was not an option and an LLM based methodology was used.

To make the LLM classification run efficiently for both cases above, all product and service revenue codes were grouped based upon the business line from which they originated. Then the LLM classification to corresponding North American Industry Classification System (NAICS) industry and codes was run via the LLM few-shot prompting query based on the product and service descriptions and the business lines in which they belong. For some products and services, their descriptions were missing. For these, the descriptions of the same sales part number (SPN) were used (which is a lower-level description than the master code). When these division chunks are completed and combined, a check was performed to determine whether there was any failed classification. In case they did, the LLM classification process was re-run.

The NAICS codes and industry descriptions are then parsed from the direct output of the LLM classification results. Based on the parsed NAICS codes and industry description, the relevant GHG emissions factors were retrieved digitally using the LLM RAG (Retrieval-Augmented Generation) technique.

The overall product and service intensity calculation workflow is shown in Figure 9: Product and Service Intensity Calculation using LLM Workflow overview

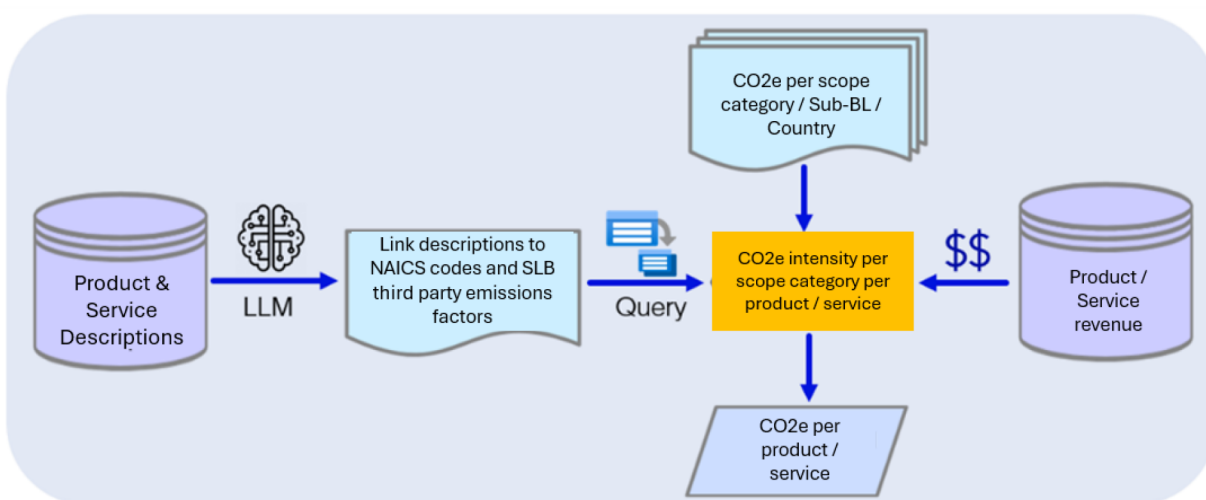


Figure 8—Product and Service Intensity Calculation using LLM Workflow overview

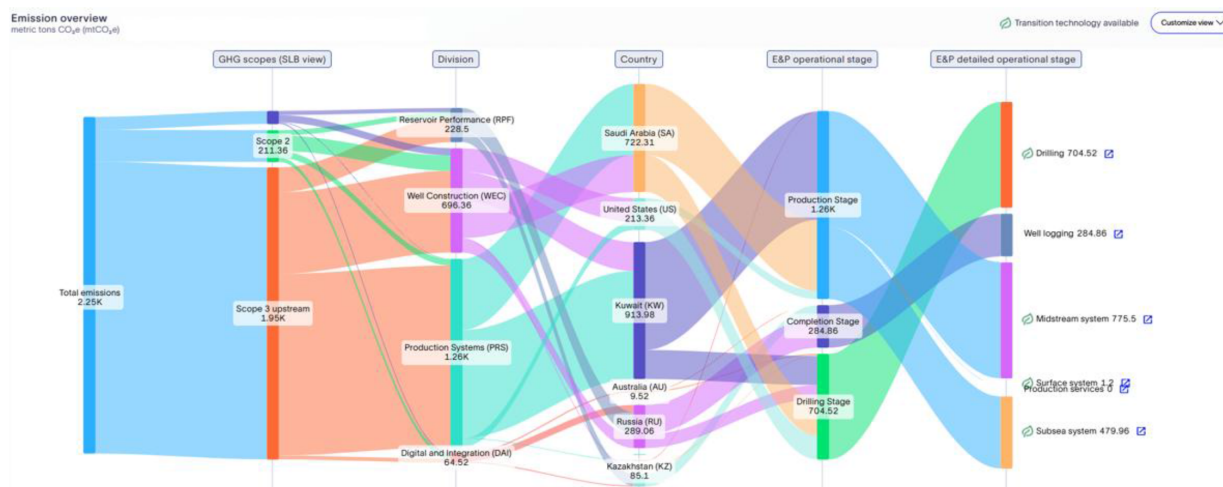


Figure 9—Example sankey view showing GHG Scope, Division, Country and E&P operational stages. Emissions data is for representational purposes only.

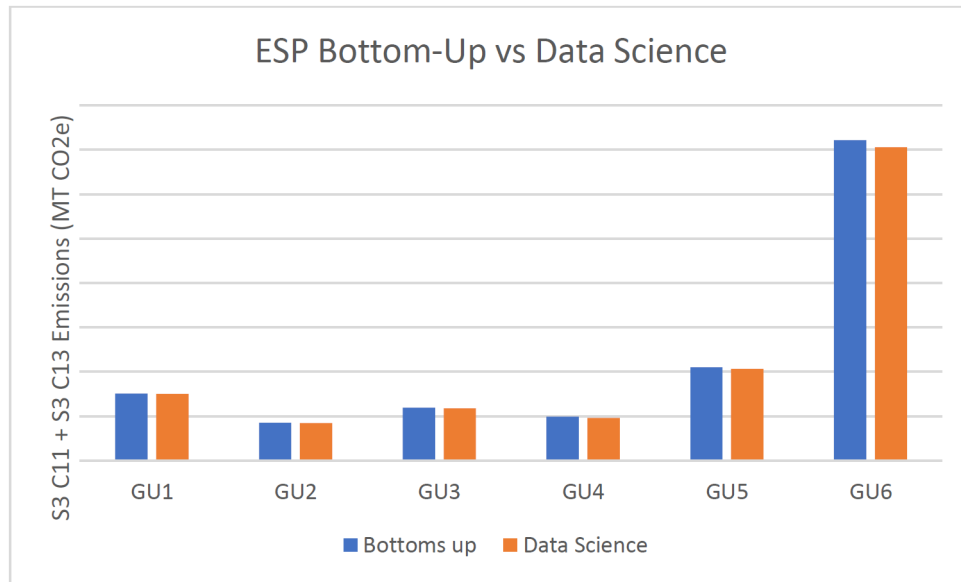


Figure 10—ESP Use Phase (S3 C11 and S3 C13) emissions comparison between bottom -up and top-down approaches

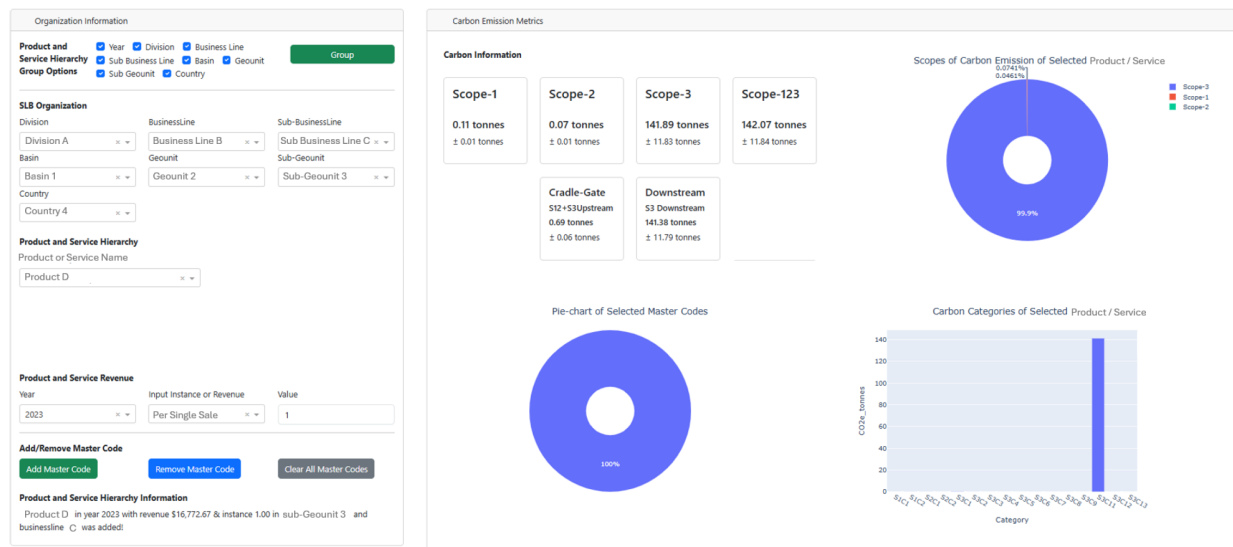


Figure 11—Product and Service Carbon Footprint Estimation, showing the estimated GHG emissions per scope category for an example product of an oil and gas service company

Once third party emissions factors are correctly matched to each product and service description, it is possible to generate a refined, company specific carbon emission factor for each product and service within the granularity unit that was previously considered for data merging and customer specific emissions calculations (CO₂e per scope category / Sub-BL / Country).

The following three equations summarize how to calculate the emissions intensities at the product and service level.

$$X[S_1 * M_1 + S_2 * M_2 + S_3 * M_3 + \dots + S_n * M_n] = CO_{2_cat} \quad \text{where } S_1 = \frac{C_1}{C_{min}}, S_2 = \frac{C_2}{C_{min}}, \dots \quad (1)$$

$$X = \frac{CO_{2_cat}}{[S_1 * M_1 + S_2 * M_2 + S_3 * M_3 + \dots + S_n * M_n]} \quad \text{where } X: \text{ base intensity} \quad (2)$$

$$MC_1 = S_1 * X, MC_2 = S_2 * X, \dots, MC_n = S_n * X \quad (3)$$

C_i: Emissions factor or intensity of a specific product or service - i,

$C_{\min}$: The minimum emissions factor intensity of all products and services contained within the specific granularity unit.

S_i : The ratio (weight) of emissions factor intensity and minimum emissions factor intensity for a given product or service. – i.

M_i : The revenue associated with each product and service – i.

CO_{2_cat} : The carbon emission of each scope category within the granularity unit.

X : The base intensity for the corresponding category within the granularity unit.

MC_i : The emissions intensity for each product or service – i.

The carbon emission for different products and/or services can be calculated once the carbon emission factors/intensities of each is available. Then the carbon emission data can be grouped in several ways which depends on the query purposes. For example, the data can be re-grouped based on the customer names, or product and service description. This product and service specific emissions data can also be used for Sankey diagram displays at a higher level of granularity.

Product and Service Emissions Statistical Uncertainty Calculation

To describe the estimated carbon emission more precisely, the uncertainty concept was introduced. The solution will provide two numbers – estimated carbon emission and uncertainty for a given product or service and corresponding revenue.

After the calculation of intensities for each category of products and services is done, the carbon emission for each sales line item of all sales orders can be achieved within the granularity. Then it is feasible to run the statistics for all products and services based on the granularity level – which can be dynamically changed by user. Besides the average carbon emission of each category for all products and services, the corresponding uncertainties are also calculated.

The uncertainties are also known as the margin of error of mean of carbon emission for each category. The following formula shows how to calculate the margin of error (ME) - uncertainty.

$$ME = t^* \times \frac{s}{\sqrt{n}}$$

For 95% confidence level, the critical value t^* depends on the degree of freedom $df = n - 1$ (n is the sample size) and the value can be found from the t-distribution table. The s is the sample standard deviation.

The uncertainty - Margin of Error represents the potential variation between the sample estimate and the actual population parameter. It indicates the degree of uncertainty associated with our sample results. A smaller Margin of Error suggests a higher level of confidence in the accuracy of our estimate. The Margin of Error is influenced by three primary factors:

- **Sample Size:** Larger sample sizes generally result in a smaller Margin of Error, enhancing the precision of the estimate.
- **Confidence Level:** The confidence level represents the probability that the true population parameter lies within the Margin of Error. Common confidence levels include 95% and 99%.
- **Standard Deviation:** This measures the variability of the data. Higher variability increases the Margin of Error.

Results

The entire workflow was successfully performed using corporate metrics and GHG emissions data over a five year period, giving insights into the emissions contribution for different business line and geographical regions as indicated in [Figure 10](#) (note that data shown is not related to a real customer and is for representative purposes only). The Sankey diagram is beneficial in showing the flow of emissions between

different organization parameters and emissions scope in a single view that can be quickly used to identify the parts of the business that contribute the most to the GHG emissions inventory.

For data validation we compared products and services for which we have developed bottom-up emissions calculation methods with our top-down approach. One example of this is estimated lifetime GHG emissions for electrical submersible pumps (ESPs). The comparison between the top-down data (using the data science approach explained in this paper) for Scope 3 Cat 11 and Scope 3 Cat 13 emissions from ESPs is shown below for the top 6 highest emitting geounits in 2023. Top-down data is calculated using the approach described in this paper whilst the bottom-up approach is computed for each ESP sold using its expected lifetime GHG emissions, the electrical grid intensity in their location, the motor power for the specific ESP and the expected uptime during its run life. As can be seen in Figure 12: ESP Use Phase (S3 C11 and S3 C13) emissions comparison between bottom -up and top-down approaches agrees extremely well with the average difference being just 1.6%.

A digital interface was built to visualize the emissions intensity from individual products, using the revenue-based approach shown in the methodology section. The interface can give an estimation of the product and service specific GHG emissions across different scopes, and can output a location specific approximation of revenue per unit product sold or service rendered – hence enabling the estimation of product or service carbon footprint based upon unit quantity sold. See Figure 12: Product and Service Carbon Footprint Estimation for an example of this output:

Discussion

By integrating corporate greenhouse gas (GHG) inventories with enterprise resource planning (ERP) data—including revenue and product/service information, the approach enables an estimation of emissions at the product and service level. Statistical imputation techniques address emissions missing data by utilizing distribution characteristics across business lines and geographic units, allowing for consistent assignment of emissions values when direct data are unavailable. Collectively, these methods yield over 560,000 emissions intensity factors, each specific to business organizational unit, geographical organizational unit, emissions scope category and time period. Results demonstrate a good correlation between these top-down, data-driven estimates and bottom-up, process-specific emissions calculations for selected products. Furthermore, digital visualization tools enable rapid exploration of emissions data, supporting the identification of trends, emissions hotspots, and decarbonization opportunities across multiple organizational levels. The methodology presented advances in the calculation and interpretation of product- and service-level carbon footprints, providing a basis for more targeted and effective emissions reduction initiatives.

Conclusions

The scalable data science approach presented in this paper offers a significant advancement in estimating carbon footprints for products and services in the oil and gas sector. By integrating corporate emissions data with enterprise business metrics, the methodology enables granular insights into greenhouse gas (GHG) emissions at the product and service levels. This approach addresses the challenges of traditional manual life cycle assessments, which are often infeasible at scale due to the complexity and geographic variability of oil and gas operations.

The use of statistical imputation techniques ensures comprehensive emissions allocation across organizational units, even in the presence of missing data. Additionally, the calculation of carbon intensities for each scope category, business line, and geographic unit provides a robust framework for estimating customer-specific emissions based on associated revenues.

Leveraging large language models (LLMs) can significantly streamline and enhance our ability to determine product- and service-specific emissions. LLMs enable automated classification and mapping of products and services to relevant emissions intensity metrics, which improves both efficiency and accuracy

in our methodology. However, for this process to fully realize its potential, we require a more granular dataset—one that directly links detailed product or service attributes to greenhouse gas emissions. Without such a dataset, the precision of our emissions estimates for individual products and services remains limited, even with advanced automation.

The successful validation of this methodology using actual oil and gas service company data from 2019 to 2024 demonstrates its practical applicability and reliability. The detailed emissions insights, visualized through Sankey diagrams, show strong agreement with bottom-up emissions estimates, exemplified by the case of electrical submersible pumps.

In conclusion, this data-driven approach provides a scalable and accurate solution for estimating product carbon footprints in the oil and gas sector, supporting targeted decarbonization efforts and contributing to cost effective solutions to help reduce emissions in the oil & gas industry.

The methodology also shows promise to be applicable beyond oil and gas, to other sectors involving a complex array of products and services in which traditional methods of computing product carbon footprint is challenging. So long as a company has enough enterprise data to build statistically relevant emissions factors with sufficient and meaningful granularity, the method can be adapted to help them gain insights into the carbon footprint of their products and/or services.

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 - Suggesting relevant information to be included in the introduction section
 - Clarifying the written content of multiple parts of the report, after initial drafts by the author.
- All AI generated content has been validated by the author*

Definitions

- Product Carbon Footprint (PCF)** : The total greenhouse gas (GHG) emissions associated with a product throughout its entire life cycle, from raw material extraction to disposal
- Greenhouse Gas (GHG)** : Gases that trap heat in the atmosphere, contributing to global warming and climate change. Common GHGs include carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O).
- Large Language Model (LLM)** : Advanced machine learning models that use vast amounts of data to understand and generate human-like text. In this context, LLMs are used to automate the classification of business activities and map products and services to emissions intensity indicators.
- Carbon Intensity** : The amount of greenhouse gas emissions produced per unit of economic activity, for example emissions per unit of revenue.
- Statistical Imputation** : A method used to estimate missing data points by using statistical techniques to predict values based on existing data.
- Sankey Diagram** : A type of flow diagram that visualizes the distribution and flow of resources, such as emissions, within a system. It is used to illustrate the relative magnitude of different pathways.
- Scope Categories** : As defined within the GHG protocol, classifications of greenhouse gas emissions are based on their sources. Scope 1 is defined as emissions from operations that are owned or controlled by the reporting company. Scope 2 is defined as emissions from the generation of purchased or acquired electricity, steam, heating, or cooling

- consumed by the reporting company and scope 3 includes all indirect emissions (not included in scope 2) that occur in the value chain of the reporting company, including both upstream and downstream emissions.
- Oil Production Greenhouse Gas Emissions Estimator (OPGEE) Life Cycle Assessment (LCA)** : (Brandt, A. R., et al. (2022)) A model developed to estimate the greenhouse gas emissions associated with different stages of oil production.
- Decarbonization** : The process of reducing carbon dioxide emissions by using low-carbon power sources, increased energy efficiency, and other measures aimed at reducing the carbon footprint.
- Division / Business Line / Sub Business Line** : Definition of business organization hierarchy presented in this paper. Each division has multiple business lines and each business line has multiple sub-business lines.
- Basin / Geounit / Sub-geounit / Country** : Definition of geographical organization hierarchy presented in this paper. Each basin has multiple geounits and each geounit comprises multiple sub-geounits and countries.

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